AXON BLUETOOTH VULNERABILITY

A Mitigation for Law Enforcement

A Sovereign Counter-Surveillance Capability for the BLE-OUI Tracking Threat

CONFIDENTIAL — RESPONSIBLE DISCLOSURE

PROJECT NULLWEAR

Selective per-packet annihilation, deployed per-officer.
A companion to the prior disclosure report on the underlying vulnerability.

Prepared by

Benjamin Jack Leighton

Tester Present Specialist Automotive Solutions

Date of Issue

06 May 2026

Classification

CONFIDENTIAL — Restricted to authorised disclosure recipients

This document accompanies a prior disclosure report describing a real-world weaponisation of the underlying vulnerability. It is intended solely for relevant law enforcement, national-security and regulatory recipients under responsible disclosure. Unauthorised distribution or operational use of any technique described herein is strictly prohibited.

1. Executive Summary

This report sets out a complete, fundable, deployable mitigation strategy for the Axon Bluetooth Low Energy (BLE) tracking vulnerability previously disclosed by the author. The vulnerability — that all Axon Enterprise law-enforcement equipment broadcasts a fixed, manufacturer-tied identifier (MAC OUI **00:25:DF**) which can be passively detected at ranges of up to ~400 m by any commodity Bluetooth receiver — has now been demonstrated to be weaponisable at city scale by a single attacker with under USD 100 of hardware per scanner node and a commodity cloud account.

This document proposes a sovereign Australian counter-capability that addresses the vulnerability without requiring cooperation from Axon Enterprise, without new legislation, without infrastructure rollouts, and without changes to the operational behaviour of officers in the field.

The capability is designated **PROJECT NULLWEAR**. Its central component is a coin-sized, officer-issued radio device that performs **selective per-packet annihilation** of the officer's own Axon BLE broadcasts, in flight, at the radio layer, before any external scanner can receive a clean copy. The technical result is that an officer wearing the device is **invisible** to any third-party BLE-OUI tracking system — anywhere in the country, in any setting, regardless of attacker scanner density, regardless of suburban or industrial coverage gaps, and regardless of whether Axon ships any firmware update in the next decade.

Around the central capability, four supporting layers harden the defence against edge cases and against attacker adaptation: covert-vehicle decoy emission for plainclothes operations; AFP-operated cloud-side poisoning of identified attacker backends; telco-side fingerprinting and physical recovery of attacker scanner hardware; and long-term procurement pressure on Axon Enterprise to ship Resolvable Private Address (RPA) firmware as a default configuration.

Bottom line: for an indicative federal expenditure of approximately **AUD 7 million in hardware** and a **90-day deployment window**, every Axon-equipped officer in the country can be made structurally invisible to the entire class of distributed BLE-OUI-based police-tracking systems, with no operational disruption to existing equipment, no change to officer behaviour, and no dependency on any third party.

2. Scope and Relationship to Prior Disclosure

This report is a companion to, and operational complement of, the prior disclosure report titled *Weaponised Bluetooth Tracking of Law Enforcement Personnel*, issued by the same author under the same confidentiality regime. The prior report describes the vulnerability, the recovered weaponisation, and the threat model. This report describes **what to do about it**.

The two documents should be read together. The reader is assumed to be familiar with the underlying vulnerability — namely, that Axon BLE-emitting equipment is identifiable on-air at range by the OUI in its broadcast MAC address, and that a productionised system has been recovered which exploits this property at city scale via a distributed mesh of independent scanner nodes feeding a central cloud-hosted dashboard.

Prior disclosure history. The underlying vulnerability was first formally disclosed to Victoria Police in late 2022 by the author, predating the public DEF CON 31 (2023) talks on the same subject. The present mitigation report is therefore not a first-time disclosure of the vulnerability; it is the operational answer to the question that disclosure has been asking for three years.

3. Why the Obvious Mitigations Fail

Any serious mitigation proposal must first explain why the natural first answers are wrong. Five categories of "obvious" countermeasure have been considered and rejected for the reasons set out below.

3.1 Broadband BLE jamming

Filling the 2.4 GHz band with noise has three fatal problems. (i) It is illegal under the Radiocommunications Act 1992 (Cth) and equivalents in every Western jurisdiction. (ii) 2.4 GHz is shared by Wi-Fi, Zigbee, medical telemetry and the police's own BLE equipment — a jammer would brick body cameras, holster sensors and Taser pairing along with the attacker's scanners. (iii) A jammer is detectable on-air; the attacker's mesh would simply log "coverage dropped during interval Y" and route around the jammer's footprint. Wrong tool.

3.2 Faraday-shielded equipment carriers

Physically possible, commercially available, but operationally hostile. Axon's product line depends on BLE being live — Signal Sidearm needs the holster to talk to the body camera, the camera needs to pair with a phone or controller, Taser telemetry needs to reach Axon Aware. Shielding kills all of that. Per-officer cost is high; usability cost is higher.

3.3 Disabling BLE on Axon equipment

Same fundamental problem as shielding. A "BLE off" mode is useful for covert / undercover work but cannot be the default state of an active patrol officer's equipment without losing the operational features that justify the equipment's price.

3.4 Lawful takedown of attacker cloud backends

Necessary but not sufficient. The defender can subpoena the cloud provider, seize the tenant, and preserve evidence — but the attacker spins up a new tenant on a different provider in twenty minutes. This is a whack-a-mole posture and the attacker has the structural advantage of being the one who chooses when and where to deploy. Useful as a sustained-pressure supporting layer; not the answer.

3.5 Waiting for Axon to ship Resolvable Private Address (RPA) firmware

This is the engineering-correct fix at the protocol layer, and it is the long-term answer. It is not the near-term answer. Axon will take years to ship RPA across all SKUs; the change may break Signal Sidearm pairing logic that depends on a stable MAC; some fielded hardware may not support RPA without a hardware refresh; and every device already deployed in the field stays vulnerable until physically replaced. This track must be pursued in parallel — see §7.5 — but it is not what protects officers in 2026.

3.6 Decoy-only mitigations

Adding fake 00:25:DF beacons to the radio environment — whether per-officer, per-vehicle, per-utility-cabinet or otherwise — does **degrade** the attacker's signal but does not **eliminate** it. Decoys can be filtered with sufficient observation time (via correlation, RSSI stability analysis, MAC co-occurrence, location clustering against known infrastructure maps, and so on). More importantly, decoys leave the attacker fully operational in coverage gaps — quiet suburban streets, industrial estates, regional towns and rural roads where infrastructure decoy density is necessarily lower. **An attack class that operates everywhere cannot be defeated by a defence that operates only somewhere.** Decoys are a useful supporting layer (see §7.2) but cannot be the primary defence.

4. The Strategic Insight — Annihilate, Don't Mask

The conceptual breakthrough is to stop trying to manipulate the attacker's perception of the signal and instead to **destroy the signal at its source, in the air, before it crosses the perimeter of a protected zone**.

> Decoys add noise to the attacker's input. Noise can be filtered. Annihilation removes the signal from the attacker's input. There is nothing left to filter. The two strategies are categorically different. Decoys make the attacker uncertain. Annihilation makes the attacker blind.

The technical means of doing this are well-characterised in the open academic literature. BLE advertising is broadcast on three known channels (37, 38, 39), in packets of a known structure, at intervals that fall within a known range. A purpose-built device — the size of a 50-cent coin and costing in the order of AUD 50 in production — can:

- Continuously listen across all three BLE advertising channels.
- Detect, in real time and within microseconds of the packet preamble, whether the incoming packet's MAC field begins with 00:25:DF.
- Immediately emit a precisely-timed colliding pulse on the same channel during the packet's CRC trailer.
- Cause every BLE receiver in range — including the attacker's scanner mesh — to fail CRC validation and silently discard the packet.

To the attacker's mesh, an officer wearing this device is **not detected at all**. The attacker sees no packet, not a corrupted packet — because BLE receivers discard CRC failures without logging them. This is the qualitative shift the previous, decoy-only proposals lacked.

The technique is described in published, peer-reviewed academic work — see for example Heinrich et al. on selective denial-of-service for BLE advertisements, and Brauer et al. on reactive jamming on Bluetooth Low Energy. It has been demonstrated using off-the-shelf hardware costing under AUD 30. Its application to an officer-protection use case is novel; its underlying engineering is not speculative.

5. NULLWEAR — The Core Capability

This section describes the proposed sovereign capability in operational terms — what is issued, to whom, in what form, with what hardware, with what behaviour, and under what legal authorisation.

5.1 Operating principle

NULLWEAR is a passive-receive / reactive-transmit BLE radio device. It carries no information, performs no encryption, transmits no advertising packet of its own, and cannot be paired with. It performs exactly one function: **upon detection of any BLE advertising packet whose MAC begins with the Axon Enterprise OUI, emit a colliding pulse during the packet's CRC region, causing the packet to be discarded by every receiver in range**.

The device's effect is bounded to its own radio range — typically 10–30 m for a body-worn unit, up to 50 m for a vehicle-mounted unit with an external antenna, and the entire interior of a building for a station-mounted unit with multiple distributed antennas. Within that range, the targeted broadcast does not exist on the air. Outside that range, every other radio service operates entirely normally.

5.2 Hardware

The reference platform is the Nordic Semiconductor nRF5340 system-on-chip, which has a dedicated radio core capable of the sub-microsecond reactive timing required for per-packet CRC corruption. The same architecture is supported by the nRF52840 (used in published academic demonstrations of the technique) and by the Texas Instruments CC2652 family. All three chips are mature, second-source-available, and cost under AUD 10 each in volume.

Form factor	Target SoC	Battery / power	Antenna	Approx. range	All-in unit cost (50k vol.)
NULLWEAR/P (Personal)	nRF5340 in matchbox potted enclosure	200 mAh LiPo, USB-C charge, ~18–24 hr/charge	Internal PCB whip	10–30 m	AUD 50–60
NULLWEAR/V (Vehicle)	nRF5340 with external front-end	Hard-wired to 12 V vehicle bus	External roof or window-glass antenna	30–50 m	AUD 70–90
NULLWEAR/S (Station)	nRF5340 cluster, multi-antenna	AC mains with battery backup	Distributed ceiling antennas	Building-scale	AUD 350–500

5.3 Firmware

The total core logic is approximately 150 lines of C running on the nRF radio core. Pseudocode form:

```
loop:
on every BLE advertising channel (37, 38, 39):
listen for packet preamble
on preamble detected:
capture first 6 bytes (access address + first MAC byte)
if first 3 MAC bytes == 0x00 0x25 0xDF:
switch radio to TX mode
emit collision pulse for ~30 us during the packet's CRC region
switch back to RX mode
```

The radio core's hardware-level packet reception is fast enough to detect the OUI within the first ~32 bits of the MAC field, leaving ~120 µs to switch to transmit and emit the collision before the CRC is sent. This is well within the chip's published radio-switching specification.

5.4 Form factors and issue

- **NULLWEAR/P** — personal, body-worn. Clipped to belt, pocketed, or integrated into the Axon body-camera mount. Battery life 18–24 hours. Charged on the same dock as the body camera, requiring no additional officer-side process.
- **NULLWEAR/V** — vehicle-mounted. Wired into the 12 V system. Larger antenna provides a ~50 m bubble. Covers the officer between vehicle and incident scene, and protects additional officers operating within range of the vehicle.
- **NULLWEAR/S** — station-mounted, AC-powered, multi-antenna. Covers the entire station building and immediate carpark. Solves the high-density problem of officers congregating at briefings, charging gear, or exchanging shifts — situations in which many real Axon broadcasts would otherwise be visible at one location at one time.

5.5 Legal classification

NULLWEAR is **not** a jammer in the regulatory sense. A jammer transmits continuous interference across a band. NULLWEAR transmits only when it detects the leading bytes of an Axon-OUI BLE advertising packet, only on the channel that packet is on, only for the ~30 microseconds it takes to corrupt the CRC, and only within the device's own short radio range. The radio spectrum at any given instant outside the per-packet corruption window is undisturbed.

In Australian regulatory terms, NULLWEAR is a **selective passive-emission device**, operating within ISM-band power and duty-cycle limits, on the same band the protected equipment is transmitting on. ACMA can authorise the device class under existing class-licence variation provisions in LIPD 2015. The legal pathway is a single Ministerial determination — not new legislation. The same statutory instrument has been used to authorise low-power active jamming of explosive-device triggers in counter-IED contexts; this is the same instrument applied to a much narrower target signature.

> **Critical legal point.** Because the device only ever transmits in response to a specific OUI in a specific advertising packet on a specific channel, and because the device's own emissions fall within standard ISM-band class-licence limits, NULLWEAR does not interfere with any other licensed service, does not deny service to any general user, and does not constitute unauthorised access to any computer system. Its legal posture is substantially cleaner than that of a conventional jammer.

6. Why This Changes the Game — Strategic Asymmetry

The decisive property of NULLWEAR is the inversion it imposes on the attacker's economic model. The threat described in the prior disclosure report works because the attacker can deploy cheap scanner nodes and aggregate their detections into a real-time dashboard. NULLWEAR does not stop the attacker deploying scanners; it makes the data the scanners produce worthless.

Property	Before NULLWEAR	After NULLWEAR
Attacker capital required for metro mesh	~AUD 500k	Same — but now produces no useful signal
Attacker effective signal	Real-time officer location	None — annihilated at source
Attacker false-negative rate against officers	Near zero	Near 100%
Attacker dashboard utility	Fully operational	Useless
Defender hardware required	National infrastructure (~AUD 15M)	Per-officer only (~AUD 7M)
Defender depends on	Telcos, councils, NBN, Axon	Nothing external
Time to deploy	12–18 months	90 days
Attacker counter-adaptation path	Build better filters	None within BLE; must abandon attack class

The attacker's only response is to abandon BLE-OUI tracking entirely as a workable signal. Every alternative sensor available to them — visual identification, automatic licence-plate recognition, social engineering, human surveillance — is more expensive, slower, more detectable, and more attributable than passive BLE listening. NULLWEAR forces the attacker out of stealth and into modalities where law enforcement has long-established detection and prosecution capability.

7. The Defence-in-Depth Stack

NULLWEAR is the primary defence. Four supporting layers harden it against edge cases the personal annihilator alone cannot address.

7.1 Layer 1 — NULLWEAR (annihilation)

Already described in §5. This is the load-bearing layer; the others exist to back it up.

7.2 Layer 2 — Covert-vehicle decoy emission

Officers operating in plainclothes or undercover roles cannot wear an actively-radiating device. For these officers, NULLWEAR's protection model is inverted: rather than annihilating the officer's own broadcast, a small, low-power decoy emitter is installed in the covert vehicle and broadcasts a single static fake 00:25:DF MAC continuously.

To the attacker's mesh, the covert vehicle therefore appears to be the location of an Axon device **at all times**, regardless of whether an officer is actually inside it. The attacker's live-information signal is buried in a continuous false signal; their ability to infer officer presence from BLE detection is destroyed for that vehicle.

Per-vehicle hardware cost: under AUD 20. Total covert-fleet cost across all jurisdictions: well under AUD 1 million.

7.3 Layer 3 — AFP cloud-side poisoning of identified attacker backends

The attacker's mesh sends every detection to a single cloud backend. The recovered code shows that the backend has no authentication on its ingestion endpoints. The Australian Federal Police, under appropriate authority, can operate a standing daemon that — given a known attacker backend URL — submits high-volume fabricated detection traffic from controlled cloud infrastructure.

This poisons the attacker's database independently of the airwave defence. Their dashboard is inundated with millions of plausible but false detections per second. Their cloud bill spikes. Their database fills. Even any signal that somehow reached them despite NULLWEAR is buried in cloud-injected noise.

This capability runs continuously, attacking the same problem at two layers — radio and cloud — neither of which the attacker can defend against without abandoning their entire architecture.

7.4 Layer 4 — Telco-side fingerprinting and physical recovery

Each attacker scanner node maintains a network-side signature. The recovered code uses a 5-second heartbeat to a fixed cloud endpoint, with low data volume, with a static SIM attached to a fixed cell tower for weeks at a time, with no voice or SMS traffic. That is a highly distinctive on-network pattern.

Coordinated Telstra / Optus / Vodafone / TPG SIEM rules can flag SIMs matching the pattern. Each match becomes a warrant target. Cell-tower triangulation produces a physical location; physical recovery follows. Each recovered device gives more signature data — enclosure type, supplier, firmware version, additional MAC fingerprints — that feeds back into the telco rules and into customs / border alerts on bulk hardware imports.

Sustained, this is a continuous stripping action. Every attacker node that gets recovered reduces their mesh by one. Every replacement node they buy gets recovered too. The attacker's mesh degrades faster than they can rebuild it, while the airwave annihilation ensures even the operational nodes are useless.

7.5 Layer 5 — Long-term vendor track for Axon RPA firmware

Independent of the above, joint procurement pressure on Axon Enterprise from every Australian, New Zealand, Canadian and United Kingdom agency operating Axon equipment can make **mandatory Resolvable Private Address (RPA) firmware** a condition of the next contract renewal cycle.

Axon's Five Eyes contracts are worth hundreds of millions of dollars per year. The cost of complying with the procurement requirement is software engineering work; the cost of not complying is losing the bloc. Within three to five years, RPA-by-default eliminates the underlying vulnerability at the protocol layer. NULLWEAR continues to operate as defence-in-depth and as protection for fielded equipment that has not yet been firmware-updated.

8. 90-Day Deployment Plan

The deployment pathway is intentionally compressed. The capability is engineered around the premise that officers are exposed today and the answer cannot be a multi-year capability build. Each milestone below is achievable with existing supply chains and existing regulatory instruments.

Phase	Days	Deliverable	Owner
Stand-up	0–14	Joint Capability stand-up: Home Affairs CICC + ASD ACSC + Victoria Police + AFP. ACMA class-licence variation issued. National procurement contract awarded for 100k nRF5340-based units (domestic manufacturer).	Home Affairs CICC
Pilot	15–45	200 NULLWEAR/P units issued to the metro central command of the participating state police service. Concurrent red-team operation: stand-up of the recovered attacker stack inside a controlled environment, deployed against pilot officers, validates that the attacker's dashboard reports zero detections of those officers despite scanner nodes within 50 m.	State police TSU + CICC
Phase-1 issue	46–60	First 10 000 production units delivered. Issued to: state police on-duty patrol officers; AFP; protective-service officers at parliamentary, court and diplomatic facilities.	Each agency procurement
National scale	61–90	Scale to 50 000 NULLWEAR/P units across all state police services, AFP, ABF and emergency-services agencies. Vehicle and station variants follow on the same supply chain through months 4–6.	Federal-state coordination

By day 90 the attack class is operationally defeated in every jurisdiction that has chosen to participate.

9. Indicative Cost

Costs below are unit-economics for hardware only. Distribution, training, integration with existing equipment lockers, and contingency are typically budgeted at +30%.

Line item	Quantity	Unit cost (AUD)	Subtotal (AUD)
NULLWEAR/P personal device	50,000	$60	$3,000,000
NULLWEAR/V vehicle device	25,000	$80	$2,000,000
NULLWEAR/S station device	600	$400	$240,000
Covert-vehicle decoy emitter (Layer 2)	3,000	$20	$60,000
Hardware subtotal			**$5,300,000**
Distribution, integration, contingency (+30%)			$1,590,000
All-in capital expenditure			**$6,890,000**

Annual operating cost is negligible: NULLWEAR has no recurring service fee, no SIM, no cloud tenancy, no licensing. Battery replacements at the personal-device level are budgeted at approximately AUD 5 per officer per year. Layer 3 (cloud-side poisoning) is operated within existing AFP cyber-capability budget. Layer 4 (telco fingerprinting) is operated within existing telco SIEM and existing AFP / ASIO investigation budgets. Layer 5 (vendor pressure) has no marginal cost.

10. Risks and Limitations

Honesty about what the proposed capability does and does not do is essential for credible adoption.

• **NULLWEAR does not protect equipment that is outside the wearer's immediate radio range.** An Axon body camera left on a desk in an unprotected room will broadcast normally and is detectable. Mitigation: NULLWEAR/S covers all stations; equipment lockers are inside that coverage by construction.

• **NULLWEAR does not address legacy detections already collected by attacker meshes.** Whatever historical data the attacker has accumulated remains usable for retrospective pattern-of-life analysis on individual officers. Mitigation: Layers 3 and 4 work to neutralise existing meshes and seize their data stores.

• **NULLWEAR depends on the attacker continuing to use BLE-OUI-based detection.** If the attacker pivots to visual identification, ALPR, social engineering or human surveillance, NULLWEAR no longer applies. Acceptable: those alternative modalities are categorically more expensive, slower, more detectable, and more prosecutable. Forcing the attacker into them is a strategic win.

• **NULLWEAR could in principle be detected by a custom SDR-based scanner that logs CRC failures.** Such a scanner is not part of the recovered system and would require a complete attacker re-engineering effort. Even if deployed, a CRC-failure log entry contains no recoverable MAC and therefore tells the attacker only "an Axon-OUI packet was being transmitted somewhere within range of this scanner" — which is true of every place a real officer or a decoy is present, and is not actionable intelligence.

• **NULLWEAR is a Australian-sovereign capability and does not protect officers in Five Eyes partner jurisdictions until those jurisdictions adopt an equivalent measure.** The design is intentionally exportable and the specification can be shared under existing intelligence-sharing arrangements; this is a feature, not a bug.

11. Recommendations

11.1 For the National Security Committee of Cabinet

• Authorise **PROJECT NULLWEAR** as a sovereign capability program with an indicative capital appropriation of AUD 7 million and a delivery target of 90 days from authorisation.

• Direct the Department of Home Affairs to convene the Joint Capability Stand-up under §8.

• Direct ACMA to issue the necessary class-licence variation in parallel with stand-up, rather than sequentially.

• Direct AFP and ASIO to operationalise Layer 3 (cloud-side poisoning) and Layer 4 (telco fingerprinting) under existing authorities.

11.2 For state and territory police services

• Identify the operational areas of greatest exposure (typically major-event policing, covert operations, witness-protection details, judicial-officer protection, organised-crime task forces) for first-issue prioritisation.

• Integrate NULLWEAR/P into the existing Axon body-camera issue process. The device is carried in the same equipment loadout, charged on the same dock cycle, and inspected at the same shift-briefing checkpoint.

• Update counter-surveillance procedures to include passive BLE-OUI tracking as a recognised threat category, and include NULLWEAR/V installation in the standard patrol-vehicle outfit.

11.3 For Axon Enterprise (the equipment vendor)

• Adopt **Resolvable Private Addresses (RPA)** as the default broadcast mode on all BLE-capable products. **However, RPA on the MAC alone is insufficient** — see the addendum recommendation below.

• **Redact or rotate the model + serial cleartext fields in the BLE Service Data 16-bit UUID 0xFE6B payload (specifically bytes 14–22 of the FE6B payload).** Empirical analysis of real-world Axon BLE telemetry (see companion *Threat Validation Report* Rev B §12.1) demonstrates that every Axon Body 3 / Body 3 Plus broadcasts its complete hardware model code and 4-character serial-number suffix in plaintext on every advertisement. An RPA fix on the MAC would still leave this stable identifier broadcast on every packet, rendering the privacy fix half-done. The complete fix requires the FE6B-payload identifier fields to rotate with the same cadence as the MAC, OR to be removed entirely from over-the-air broadcasts and limited to the secured pairing channel only.

• **Redact or randomise the 80-bit per-device identifier in bytes 2–10 of the FE6B payload**. Empirical analysis confirms this field is unique per device, broadcast in both the deployed and docked-mode FE6B variants, and constitutes a third independent stable identifier per Axon advertisement.

• **Avoid encoding agency / jurisdiction information in the MAC[3] production-batch byte.** Empirical analysis shows clear correlation between MAC[3] and deploying jurisdiction (Vic Pol, NSW Pol, WA Pol distinguishable by passive observation alone). Future production should use a randomised production-batch byte to eliminate this jurisdiction signature.

• Where a stable identifier is required for legitimate operational pairing (e.g. holster ↔ camera), pair it cryptographically rather than broadcasting it in cleartext advertising packets.

• Provide a firmware path to disable BLE advertising for officers in covert or undercover roles.

• Issue a security advisory to all customer agencies acknowledging the OUI-based, FE6B-payload-based, and serial-number-cleartext discoverability of fielded equipment so that operational risk can be assessed.

12. Distribution and Disclosure

This report is distributed under the same confidentiality regime as the prior disclosure report. The intended recipients are:

• **The relevant State Police service**

• **The Australian Federal Police (AFP)**

• **The Australian Security Intelligence Organisation (ASIO)**

• **The National Security Hotline of Australia**

• **The Australian Cyber Security Centre (ACSC / ASD)**

• **Members of the Parliament of Australia**

• **The Australian Communications and Media Authority (ACMA)** — for class-licence variation pathway.

- **Microsoft Security Response Centre** and equivalent cloud-platform abuse channels — for coordinated takedown of identified attacker tenants.
- **Bluetooth SIG** and **IEEE Registration Authority** — for awareness of the broader profile and longer-term protocol-level guidance.

The author is willing to:

- Provide a sealed technical specification of the proposed capability to authorised investigators and procurement staff;
- Brief technical staff at any of the recipients above in person or over secure video;
- Co-operate with any authorised capability development, pilot, or evaluation program arising from this proposal.

This report is provided strictly on a CONFIDENTIAL, responsible-disclosure basis. It must not be distributed beyond the authorised recipients and their immediate technical and legal advisors. Any republication, even in part, must obtain the written consent of the author.

13. Glossary

- **BLE** — Bluetooth Low Energy. Short-range wireless protocol; advertises itself many times per second on three known channels (37, 38, 39).
- **OUI** — Organisationally Unique Identifier. The first 24 bits of a MAC address, IEEE-assigned to a manufacturer. The OUI 00:25:DF belongs to Axon Enterprise.
- **MAC address** — 48-bit hardware identifier of a network or radio device.
- **CRC** — Cyclic Redundancy Check. A trailing field in every BLE packet used by the receiver to detect transmission errors. A failed CRC causes the packet to be silently discarded.
- **RPA** — Resolvable Private Address. A privacy-preserving, rotating BLE address mode specified by the Bluetooth standard, but not enabled by default on Axon equipment.
- **SoC** — System-on-Chip. The integrated radio + microcontroller silicon at the heart of NULLWEAR.
- **SDR** — Software-Defined Radio. A general-purpose radio whose behaviour is defined in software rather than fixed in hardware.
- **Reactive jamming** — the academic term for the per-packet selective interference technique on which NULLWEAR is based. Distinguished from broadband jamming by being narrowly targeted at specific packet types.
- **NULLWEAR** — the proposed sovereign Australian counter-capability described in this report. Variants: /P (personal), /V (vehicle), /S (station).
- **Axon Enterprise** — US public company; vendor of body cameras, Tasers and related law-enforcement equipment. OUI 00:25:DF.

End of Report.

Prepared by **Benjamin Jack Leighton** on behalf of **Tester Present Specialist Automotive Solutions**. Issued 06 May 2026. CONFIDENTIAL — Responsible Disclosure.